

VERBAL REASONING AND WORKING MEMORY

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A series of experiments was carried out to explore the hypothesis that a central working memory (WM) system is utilized during verbal reasoning. WM was provisionally defined in terms of two major features of short-term memory (STM): its limited storage capacity and its use of speech-coded information. The reasoning task required subjects to match (verify) as rapidly as possible a sentence of varying grammatical form with a symbolic referent. Experiments I and II studied the effect of storing an additional STM load on sentence verification latencies. As many as six items could be correctly recalled with no slowing of verification speed. Experiment III used a different procedure in which the STM items had to be articulated aloud during verification. In this case six-item STM loads slowed verification speed considerably, and did so more for the more difficult sentences. Only a small non-significant slowing of verification speed was obtained when redundant messages were articulated. Experiment IV showed that latencies were also increased by introducing phonemic similarity into the verification task. Generally the results were not fully consistent with the hypothesis of WM as a limited capacity store called upon by the verification task. Instead they supported the view that the WM is a general executive system with a limited capacity for information processing. It was proposed that the articulatory system, used in rehearsal and concerned with speech-coding, is a "peripheral" of the more central WM executive and plays a relatively minor role in verbal reasoning.

Introduction

It has been suggested that short-term memory (STM) limitations operate not only in the traditional tasks of learning, retaining and retrieving information but extend to a wide range of information-processing activities. Some authors argue that STM functions as a scratch-pad store or working memory (WM) in specific tasks such as problem-solving (Welford, 1968), concept formation (Levine, 1966), language comprehension (Anderson and Bower, 1973) and even general intellectual performance (Posner, 1965). Others propose that in addition to acting as a temporary store the WM has executive capability and is responsible for controlling and co-ordinating a wide range of mental activity (Atkinson and Shiffrin, 1971; Broadbent, 1971; Newell and Simon, 1972; Shiffrin, 1970). The present series of experiments uses a verbal reasoning task, the verification of single sentences, and demonstrates the involvement of short-term memory factors. A preliminary summary of the experiments has already been given in a more general review of our studies in working memory (Baddeley and Hitch, 1974); the present paper

aims to explore in greater depth the role of working memory in sentence verification.

There is of course already evidence which suggests that STM plays a role in complex tasks such as concept formation (Levine, 1966) and keeping-track of multi-channel displays (Kanarick and Peterson, 1969). However, in both these examples it is clear that performance at any instant is crucially dependent on memory for prior events, and to this extent the involvement of STM is almost inevitable. In the solution of a visually presented sentence verification problem where the given information is continuously available to the subject, the need to use STM is not immediately obvious.

There remains the problem of how to go about demonstrating that short-term memory factors do operate in sentence verification. STM has typically been defined in terms of two main characteristics: (i) its limited capacity for storing information, as evidenced by the limited span of immediate ordered recall (Miller, 1956) and the restricted range of the recency effect in free recall (Glanzer, 1972); and (ii) its reliance on speech-coded information (Conrad, 1962; Baddeley, 1966). Given these characteristics it was decided to test the specific hypothesis that sentence verification utilizes speech-coded information and is interfered with by a concurrent memory span task. The question of an interaction between the recency effect in free recall and sentence verification seemed far less amenable to study. In addition, there is mounting evidence that the recency effect and the memory span do not reflect a single underlying memory limitation (Baddeley and Hitch, 1974; Craik, 1971; Glanzer, Koppenaal and Nelson, 1972). The present study therefore concentrates on the question of what demands the sentence verification task makes on the span-like aspect of STM.

The particular verification task used was devised by Baddeley (1968). The material consisted of a sentence and an accompanying letter-pair presented simultaneously. The sentence described one of the two possible orders of the letter-pair, and the subject's task was to indicate as rapidly as possible whether the actual letter-pair matched this description. An example would be *A follows B—BA*, to which the correct response would be *true*. If the letter-pair in this example had been *AB*, then the correct response would have been *false*. Task difficulty was manipulated by varying the grammatical complexity of the sentence to include active, passive, affirmative and negative forms, as shown in Table I. This particular task is representative of a wide range of sentence verification tasks

TABLE I

Examples of different grammatical forms of the verbal reasoning task

Grammatical form	Example
Active affirmative	A follows B
Active negative	A does not follow B
Passive affirmative	A is followed by B
Passive negative	A is not followed by B

There was an equivalent set of sentences using the verb "precedes" instead of "follows".

(Chase and Clark, 1972; Trabasso, 1972; Wason and Johnson-Laird, 1972). Furthermore, it can claim some general validity as a reasoning task since performance is correlated with intelligence (Baddeley, 1968), and it is known to be sensitive to both environmental and speed stress (Baddeley, De Figueiredo, Hawkswell-Curtis and Williams, 1968; Brown, Tickner and Simmonds, 1969).

In Experiment I subjects were required to carry out the task either alone or whilst holding one or two items in memory. Interference would indicate that the two tasks compete for common resources.

Experiment I

Method

Subjects

Twenty-four students from the University of Sussex took part in the experiment. They were each paid 50p.

Apparatus and procedure

Thirty-two versions of the verification task were devised, comprising all possible combinations of sentence voice (active or passive), affirmation (affirmative or negative), truth value (true or false), verb (precedes or follows) and letter order (FS or SF)*. Each version was mounted on a separate slide. Three further sets of slides were used to present the memory load items. In the first set each slide contained the single letter A. In the second, each contained a single letter from the pool BCHJQRWY, the actual letter varying randomly from slide to slide. In the third set, each slide contained a different pair of letters selected at random from the same pool.

Two Kodak Carousel projectors fitted with electro-mechanical shutters were arranged to back-project onto a screen placed immediately in front of the subject. One projector was used to present the verification problems, the other to present the memory load items. The subject indicated his solution to each problem by pressing either a *true* or a *false* response key, using a separate hand for each key. Reaction times were recorded in centiseconds using a Forth Instruments Timer and were measured from the onset of the problem.

Each trial began with a verbal warning signal followed immediately by a 1-s presentation of the memory load letter(s) and then a 1-s blank interval. The sentence verification problem was presented at the end of the blank interval and remained visible until the subject responded. He then recalled the memory letter(s) vocally, except in the control condition where the single predictable letter A had been presented.

All subjects were given an initial 3-min practice session in which they worked through a printed sheet of problems. Both for practice and during the experiment proper the instructions stressed the importance of responding as rapidly as possible consistent with high accuracy. Testing was carried out individually.

Experimental design

The 24 subjects were assigned at random to two equally sized groups which received different instructions concerning input of the memory load letters. The *vocal* group were told to read the memory load letters aloud, while the *silent* group were instructed to read them silently. All subjects read the sentence verification problems silently. Size of memory load was varied as a within-subjects factor with each subject completing a block of 32 trials under each of the three load conditions. For individual subjects the order of presenting problems was the same within each block of 32 trials. A fresh random order was used for each successive subject. For half the subjects in each group the *true* response button was assigned to the dominant hand and for the remainder the assignment was reversed. Within each of these subgroups the order of presenting the memory load conditions was completely counter-balanced.

* Use of the letters SF instead of AB was purely a matter of administrative convenience.

Results

Recall of the memory items was virtually errorless for every subject, so interest can be focused on the effect of memory load on the times taken to solve the problems.

Since the experiment is concerned with the interaction between sentence verification latencies and short-term storage rather than a detailed analysis of the verification latencies themselves, the 32 problem-types were reduced to a smaller number of categories. Previous unpublished results obtained by the authors have shown that performance depends primarily on the grammatical form of the sentence and the truth value of the response. There is relatively little influence of letter order and use of the verb *precedes* rather than *follows*. Accordingly, the reasoning problems were divided into the eight basic types given by the combinations of voice, affirmation and truth value. The measure of performance on each of these eight types of problem was taken as the median of the four response latencies for that type. Error latencies, which occurred on 3.4% of trials, were discounted. The median of an even number of latencies was found by linear interpolation.

TABLE II

Mean reasoning solution times and error rates as a function of input mode and size of memory load

	Letters read silently			Letters read aloud		
	zero load	1-item load	2-item load	zero load	1-item load	2-item load
Mean solution latency (s)	3.07	3.35	3.21	3.33	3.26	3.41
Percent reasoning errors	3.6	2.1	4.2	3.4	2.1	5.0

Table II shows solution times averaged across all types of problem as a function of memory load and the mode of reading the memory items. There was no obvious effect of either memory load or reading mode. The untransformed data were subjected to a six-way analysis of variance with as between-subjects factors (i) reading mode and (ii) hand-response assignment. The within-subjects factors were (i) memory load, (ii) voice of the problem sentence, (iii) negation of the problem sentence and (iv) the nature of the correct solution (true or false). There was no main effect of memory load ($F(2,40) < 1$), nor were there any interactions involving load. In general, the only factors affecting solution times were those concerned with the form of the problem itself, though there was one marginally significant complex interaction involving reading mode and response-hand assignment.

Errors of verification were also examined (see Table II). These showed no effect of either memory load ($F(2,44) = 2.36$, $P > 0.05$) or reading mode ($F < 1$). Thus there is no evidence that subjects were managing to keep their verification latencies constant as memory load increased by allowing their error rates to rise. One can therefore be reasonably confident that memory load did not have the predicted effect on performance.

Discussion

The absence of a memory load effect suggests one of two possible conclusions: either short-term memory has nothing in common with the verification task, or, alternatively, there is some overlap but a load of two extra memory items was insufficient to show it. Since it is more cautious to accept the second possibility, a further experiment was carried out in which the memory load was made much larger. It was thought that a load of six items, which is approaching the limit set by the memory span, ought to pre-empt STM almost entirely. If a load of this magnitude still has no effect on verification latencies, the STM demands of the latter must be virtually non-existent.

Experiment II

Method

Subjects

Twenty-four Stirling University students took part in the experiment on a voluntary basis.

Apparatus and procedure

The 32 problems and the details of their presentation were the same as in Experiment I. The memory load items were lists of six consonants selected at random without replacement from the letters BHJQRTYZ. They were presented to the subject by a tape-recorder at a rate of one item/s.

Performance on the reasoning problems was studied with and without the six-letter memory load. In the *preload* condition the subject heard a verbal "ready" signal at the start of each trial followed by a list of six spoken letters. Immediately after the last letter he was presented with the sentence verification problem, which remained visible until he made his response. To complete the trial the subject attempted verbal recall of the memory items in their correct serial order. In the *control* condition the warning signal was followed immediately by the verification problem. After completing the problem the subject heard a six-letter sequence and then gave immediate spoken recall. In this way storage load during verification was manipulated while roughly equating the two conditions for total STM activity.

Each subject was given a preliminary 3-min practice session with a written version of the verification task as described in Experiment I. He then completed two successive blocks of trials, one under the control condition and the other under the preload condition. The order of conditions was counterbalanced across subjects. The order of presenting problems was the same in both conditions for each individual subject, different orders being used for successive subjects. All subjects were tested individually.

Instructions

It seemed likely that subjects' strategies might be influenced by the relative importance of the two tasks. Accordingly, different sets of instructions were given to each of two separate groups of subjects. The *equal stress* group were told to carry out the verification task as rapidly as possible consistent with high accuracy and to attempt to recall all six letters correctly. The *memory stress* group received the same general instruction, but were also told that their sentence verification performance could only be considered if they recalled all six letters correctly.

Experimental design

A split-plot design was used with two between-subjects factors (instructional emphasis and order of presenting the memory load conditions) and four within-subjects factors (memory load, voice of the sentence, presence of negation, correct response true or false).

Results

Errors of verification occurred on 6.1% of trials and are discussed in further detail below.

(a) Verification latencies

Median correct solution latencies for each of the eight types of problem were calculated in the manner described for the previous experiment.* Figure 1 summarizes the results as a function of instructions, memory load and the grammatical form of the reasoning problem. These data are collapsed over (i) different orders of presenting the memory load and control conditions, and (ii) problems having different solutions (true or false). The statistics quoted below were based on an analysis of variance in which all six factors were included.

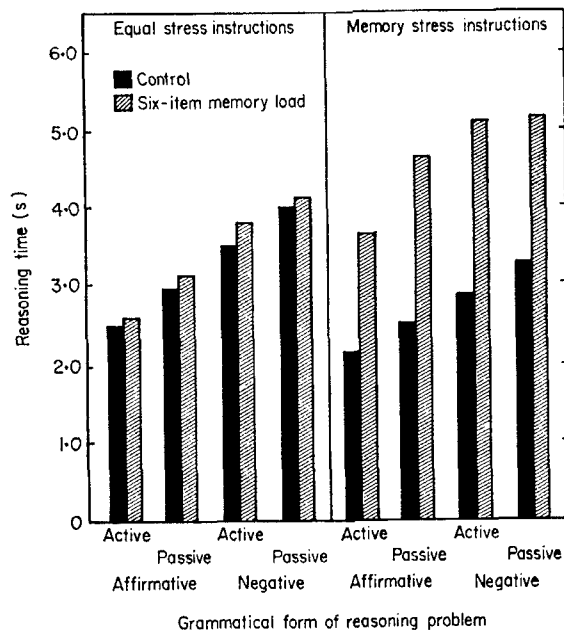


FIGURE 1. Mean reasoning solution latencies as a function of instructions and additional memory load in Experiment II.

* One point to note about the data is that reasoning times for the memory stress group in the preload condition were only scored when subjects recalled at least four out of the six items correctly. This ensured that reasoning performance was only taken into account when there was reasonable evidence of concurrent retention. The same procedure was not applied to the other group since their recall was rather too low for this to leave sufficient data. The imbalance turns out to be unimportant since reasoning times associated with recall falling below the criterion were generally longer. The scoring procedure was therefore conservative with respect to the hypothesis under test.

Additional memory load substantially interfered with verification speed for the group receiving "memory stress" instructions ($F_{(1,20)} = 16.20$, $P < 0.01$) but had no detectable effect for the "equal stress" group ($F_{(1,20)} < 1$). The interference produced by memory load with the memory stress instructions was roughly constant regardless of problem complexity. Thus, although passive problems were harder than active ones ($F_{(1,20)} = 18.95$, $P < 0.001$) and negatives were harder than affirmatives, ($F_{(1,20)} = 72.12$, $P < 0.001$) there was no interaction of memory load with either factor.

One additional effect of problem type is not illustrated in Figure I. Problems containing negatives requiring a *false* answer were easier than those requiring a *true* answer, while the reverse held for affirmatives ($F_{(1,20)} = 28.37$, $P < 0.001$). This is a fairly typical finding in sentence—referent matching tasks (Trabasso, 1972). As with the simpler effects of problem complexity its interaction with memory load was non-significant. All remaining F -ratios were non-significant, with the exception of a weak instruction $\times$ order $\times$ affirmative/negative interaction ($F_{(1,20)} = 5.65$, $P < 0.05$). A result at this level of significance may well be spurious in view of the large number of F -ratios in the analysis; and it will not be discussed further.

Inspection of the verification errors (see Table III) does not suggest any modification of the simple picture emerging from the foregoing analysis. Errors tended to be slightly more frequent in the preload condition for both groups but the differences were not significant in either case. If memory load had merely altered the setting of a speed/error trade-off on the verification task one would have expected the slower solutions to be associated with fewer rather than more errors. When broken down by problem type, errors were so infrequent that it was not possible to analyse them fully. As would be expected, there was a slight tendency for more errors to occur on problems producing long correct solution times. The error latencies were generally of the same order of magnitude as those of correct responses.

TABLE III

Reasoning error-rates and recall scores for the "equal stress" and "memory stress" instructional groups

	"Equal stress" instructions		"Memory stress" instructions	
	Control	Memory preload	Control	Memory preload
Percent reasoning errors	7.8	10.2	2.9	3.4
Mean no. of items recalled (max=6)	5.5	3.7	5.8	5.0

There was one odd feature of the error data that deserves comment. This was that errors were more frequent under equal stress instructions (by a Mann-Whitney test: $N_1 = N_2 = 12$, $U = 34$, $P < 0.05$). It is possible that the memory stress instructions were generally more daunting to the subjects and they adopted

a generally stricter criterion. However, their solution latencies in the control condition were not slower than those of the other group, which argues against this interpretation.

(b) Recall

The recall data are summarized in Table III. They were subjected to a three-way analysis of variance with memory load, instructions and the order of preload and control conditions as factors. For both sets of instructions, recall was better in the control condition than in the preload condition where sentence verification filled the retention interval ($F(1,20) = 150.00, P < 0.001$). This is consistent with the standard effect of delay on retention since recall in the control condition was always immediate. In the preload condition, the memory stress subjects recalled more than the equal stress subjects, whose average solution latency and therefore recall delay was shorter. This result appeared as a significant main effect of instructions ($F(1,20) = 17.74, P < 0.001$), and an instructions $\times$ load interaction ($F(1,20) = 21.20, P < 0.001$). On a simple trace decay interpretation one would have expected the memory stress subjects to show poorer rather than better recall. No other F -ratios approached significance.

(c) Relation between recall and verification latency

If two simultaneous tasks compete for a common pool of limited capacity there ought to be some tendency for good performance on one to be associated with poor performance on the other over a series of trials. The question of a dependence between recall accuracy and verification speed therefore has bearing on the working memory hypothesis. Figure 2(a) shows recall for the two groups of subjects as a function of the latency of the intervening verification response. Data for different types of sentence have been plotted separately and the solution latencies are blocked by 1-s intervals. The individual points are based on unequal numbers of observations because the various types of problem had different mean latencies, so the graphs must be interpreted with some caution. However it is reasonably clear that recall declines with increasing latency for both groups of subjects. This is the opposite to a trade-off between the two tasks. In seven of the eight cases plotted the Spearman correlations between recall and verification latency were strongly negative (see Table VI). For the eighth, involving equal stress subjects verifying passive negative sentences, the correlation was weakly positive.

The same data can be considered rather differently by regarding verification latency as the dependent variable (Fig. 2(b)). The correlations are again mainly negative but a new aspect of the results is apparent. On trials when subjects given equal stress instructions showed by their recall that they had stored all six items, their verification latencies were shorter than the average values in the zero-load control condition. This observation confirms the exact opposite of a trade-off between the two tasks.

Discussion

The interaction between short-term retention and sentence verification depends quite clearly on the importance assigned to each task. With equal emphasis on each, verification speed was unaffected by additional retention. When the recall

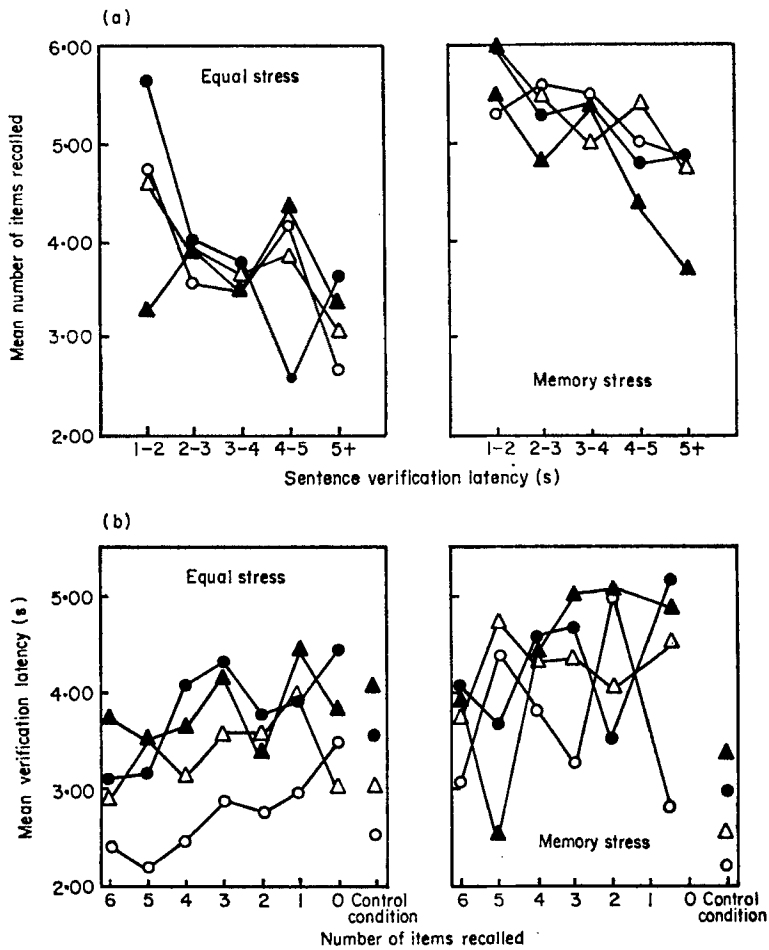


FIGURE 2. (a) Recall of the memory preload as a function of solution latency and (b) solution latency as a function of recall for different types of problem in Experiment II (○, active affirmatives; ●, active negatives; △, passive affirmatives; ▲, passive negatives).

TABLE IV

Spearman rank correlation coefficients between recall of the preload and verification latencies in Experiment II

Type of sentence	"Equal stress" instructions	"Memory stress" instructions
Active affirmative	-0.70	-0.70
Passive affirmative	-0.90	-0.90
Active negative	-0.90	-0.80
Passive negative	+0.30	-0.90

task was given the greater importance, verification was slower and recall was remarkably good. The implications for the working memory hypothesis are best considered separately for the two types of instruction, since there is good evidence that they produced rather different types of dual-task strategy.

(a) Memory stress instructions

All members of the memory stress group reported using the simple strategy of quickly rehearsing the memory items *before* attempting the verification task on preload trials. The increase in verification latencies, which was roughly constant across sentence types, appears to reflect the amount of time spent in consolidation prior to initiating the verification processes. The high recall shown by these subjects is in one sense counter-intuitive since their increased latencies lengthened the delay before recall as compared with the equal stress group. However, prior consolidation would of course tend to transfer the items to long-term memory rendering them less sensitive to delay. The evidence from this group is therefore fully consistent with their reported strategy. Unfortunately their data cannot be used as a test of the working memory hypothesis since the effective STM load during verification was probably very much lower than the nominal input. The reason for the strategy of separating the two tasks remains unclear. It could indeed be that verification requires short-term storage capacity, or that the two tasks compete for the articulatory system used in rehearsal. It may even have been a bias created by the precise wording of the instructions.

(b) Equal stress instructions

None of the subjects given equal stress instructions reported tackling one task before the other. Since it seems plausible that the memory preload was held in short-term storage, their data provide a good test of the working memory hypothesis. Yet it was this group who showed no significant increase in verification times due to extra memory load. Further on trials when all six items were correctly recalled and storage load was therefore maximal, verification times were fastest (Fig. 2(b)). This latter observation would seem to go against the hypothesis that the verification task makes demands on short-term storage. However, the general decline in recall with increasing verification latencies (Fig. 2(a)) suggests that trace-decay may be a confounding factor. Just as much as storage load may influence verification latencies, the latencies themselves may influence recall. The clearest test of the WM hypothesis is to control for any effects of delay on recall by examining retention at each delay as a function of the complexity of the sentence verified. A grammatically complex sentence verified in a given time ought on balance to require more storage capacity than a simpler one solved in the same time. Inspection of Figure 2(a) at the various solution latencies does not give a clear picture either for or against the hypothesis. Indeed in the whole experiment the only evidence in its favour is the weak positive correlation between recall and verification speed obtained with the most complex (negative passive) sentences. Perhaps a trade-off between the two tasks was in this case sufficient to offset the effects of delay on recall, which were evident with the other types of sentence. However,

this relation alone is hardly sufficient to affect the general conclusion that the short-term storage demands of the verification task did not show up as should have been expected if the WM hypothesis were true.

A common feature of Experiments I and II is that the subject was free to carry out the two tasks in alternation. This was clearly the case in the memory stress condition, and even in the equal stress condition it is very possible that subjects were using a strategy of solving the reasoning problem first, and only then attempting to retrieve the letters, with recall depending on delay rather than sentence complexity *per se*. Such a view does of course imply short-term storage independent of the WM system, and could not be regarded as consistent with the simple WM concept with which we began. We were at this point faced with the choice of either attempting to explore the task as it stood, or attempting to constrain the subjects' strategy more closely. We opted for the latter, by making two changes. First, we abandoned the preload technique in favour of a concurrent load procedure in which the subject is required to continue to rehearse the memory load items aloud whilst verifying the sentence. This technique allows separation of effects due to rehearsing the memory items from articulation *per se*. Conditions can be included in which the subject's rehearsal activity involves virtually no short-term storage so that any interference must be purely articulatory. One might expect some interference due to articulation itself since Hammerton (1969) and Peterson (1969) have previously shown effects of concurrent articulation on reasoning.

However, the crucial question here is whether memory load produces interference over and above that which is attributable to articulatory activity alone. Secondly we used digits rather than letters as the memory items, since these seemed less likely to produce specific item interference with the letters in the reasoning test, and hence might present a situation in which the subject had less need to keep the two tasks separate.

Experiment III

Method

Subjects

Subjects were 12 Stirling University students who took part in fulfilment of a course requirement.

Apparatus and procedure

All details concerning the construction and presentation of the verification task were as previously described. There were four main conditions. In the control condition each trial began with a verbal warning signal and an instruction to "say nothing". The verification task was then presented and was solved as quickly as possible consistent with high accuracy. In the second condition Murray's (1967) articulatory suppression procedure was used. After the warning signal subjects were instructed to say the word "the" repeatedly at a rate of between four and five utterances. Care was taken to ensure that they continued to articulate at the same high rate until a solution was indicated. The third condition used a procedure adopted by Peterson (1969). Here the subject was instructed to repeat the familiar counting sequence "one-two-three-four-five-six" cyclically, again at a rate of between four and five words/s. In the fourth condition the subject heard a spoken sequence of six random digits at the start of each trial and was required to repeat

the sequence cyclically at the same rate as in the other conditions. In this condition alone the message to be articulated was changed from trial to trial. The three articulation conditions therefore ranged from the repetition of a single word, through the slightly more complex counting sequence, to digit-span repetition. The latter presumably makes by far the greatest short-term storage demands. However, in terms of amount of articulation the three conditions ought to be roughly equivalent.

Subjects were each given a 3-min practice session on a written version of the verification task. They were also given preliminary practice on each of the four main conditions of the experiment.

Design

Each subject served in all four articulation conditions. The main analysis of verification performance was based on a $4 \times 2 \times 2 \times 2$ repeated measures design with articulation, sentence voice, affirmation and truth value of the solution as factors.

The 32 problems were randomly ordered within each of four blocks of trials. Within each block the four articulation conditions occurred equally often and at random, and the four blocks were so arranged that every subject completed each problem once under each of the four articulation conditions. A Latin Square was used to balance the order of assignment of articulation conditions to problems across subjects.

Results

Mean solution times and error rates for the four conditions of articulatory activity are shown in Table V. The frequency of errors did not change appreciably across conditions ($F(3,33) = 2.40$, $P > 0.05$) but solution times were affected ($F(3,33) = 14.23$, $P < 0.01$). Orthogonal comparisons showed that verification latencies were longer when random digits were recited than in the other three conditions taken together ($F(1,33) = 39.03$, $P < 0.01$). Articulation of a redundant message (counting or saying the-the-the) appears from the table to produce some interference, but the amount just failed to reach significance ($F(1,33) = 3.52$, $0.05 < P < 0.10$). This comparison may well be slightly conservative, since Wilcoxon tests showed significant interference due both to saying the-the-the ($N = 12$, $T = 14$, $P < 0.025$ one-tailed) and counting ($N = 12$, $T = 2$, $P < 0.01$). There was no difference between the effects of counting and repeating the word *the* ($F(1,33) < 1$).

TABLE V

Mean reasoning times and error rates as a function of concurrent articulatory activity

	Concurrent articulation			Random 6-digit number
	Control	"the-the-the"	"one-two-three"	
Mean solution latency (s)	2.79	3.13	3.22	4.27
Percent reasoning errors	8.1	10.6	5.6	10.3

Figure 3 shows how mean solution times for different versions of the verification task were affected by concurrent articulation. As in Experiment II, problems containing passive sentences were more difficult than those containing active

sentences ($F(1,11) = 55.13$, $P < 0.01$), and negatives were harder than affirmatives ($F(1,11) = 38.50$, $P < 0.01$). Though not shown in the figure, the interaction between affirmation and type of response (true or false) took the same form as in the previous experiment and was again significant ($F(1,11) = 16.55$, $P < 0.01$). The figure shows that the effect of concurrent articulation increased with increasing grammatical complexity of the sentence to be verified. There were significant two-way interactions between articulation and sentence voice ($F(3,33) = 5.59$, $P < 0.01$) and between articulation and negation ($F(3,33) = 5.29$, $P < 0.01$). Selected comparisons confirmed that in each case the interaction was due to the random digits condition (for the voice interaction, $F(1,33) = 15.79$, $P < 0.01$; for negation, $F(1,33) = 14.59$, $P < 0.01$). The contribution of the other three conditions was negligible ($F < 1$) in both cases. One further aspect of the results is of interest. There was no interaction between concurrent articulation and the third aspect of problem difficulty, namely the negation $\times$ type of response interaction ($F(3,33) = 2.10$, $P > 0.10$). Thus, the interference appears to have been associated with the components of the verification task particularly concerned with interpreting passive and negative sentences.

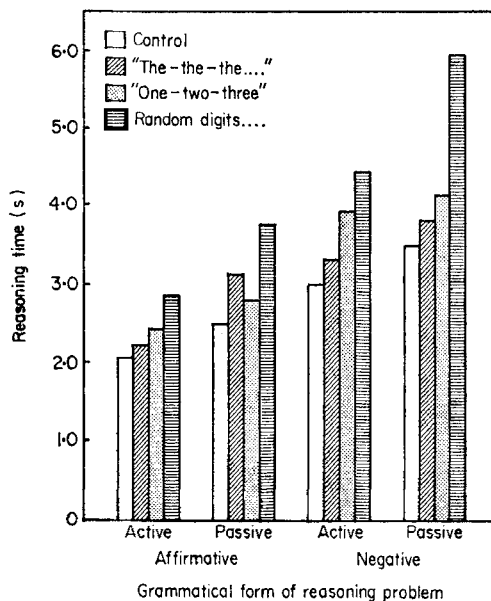


FIGURE 3. Effect of types of concurrent articulation on mean reasoning solution latencies in Experiment III.

Inspection of the error data relevant to Figure 3 suggested that there was a positive relation between solution times and error rates. The correlation was significant when computed across the 16 conditions shown in the figure ($r = 0.53$, $t(14) = 2.33$, $P < 0.05$). This provides a useful check against the suggestion that the changes in latency merely indicate chances in speed-error trade-off.

Discussion

The results suggest that rapid articulation of well-learned or simple messages has little effect on verification speed. The verification task therefore makes at most minor demands on the articulatory system. However, articulation of a novel six-digit sequence slowed verification considerably and did so more for the grammatically more complex sentences. The STM and verification tasks appear to compete for a pool of limited capacity since it is reasonable to assume that the complicated grammatical transformations take up extra capacity while the STM task occupies a roughly constant amount. Recall was unfortunately not recorded in detail so it was not possible to test for a trade-off between the two tasks on a trial by trial basis.

The interference is in contrast to Experiment II where perfect recall of a six-item STM preload was not associated with any slowing of verification. An important difference between the two procedures is that the preload technique does not force the subject to process the two tasks simultaneously. It appears then that processing operations associated with STM storage rather than storage *per se* are critical in producing interference. Implications for the WM hypothesis are dealt with more fully in the final discussion after reporting the last experiment of the present series.

Although the effect of articulatory suppression on verification speed was slight and failed to reach significance, Hammerton (1969) has reported a reliable effect with the same task. Experiment IV investigates the role of articulatory coding further by examining the effects of phonemic similarity on verification. This kind of similarity has a marked effect on the memory span (Conrad and Hull, 1964; Baddeley, 1966), and it is also found that intrusion errors in the span task are predominantly phonemically-based (Conrad, 1962; Wickelgren, 1965). In contrast, the role of visual coding in the span situation is either non-existent or weak (Laughery and Harris, 1970). Comparable effects of similarity on the reasoning task would support the view that it shares a common component with the memory span. The effects of varying both the phonemic and visual similarity of the letter-pairs used in the problems were therefore studied.

Experiment IV

Subjects

Thirty-two Sussex University students served as subjects and were each paid 50p.

Apparatus and Procedure

Nine test sheets each containing 64 verification problems printed in random order were used as material. Each of the 32 forms of the problem was represented twice per sheet. A different letter pair appeared on each sheet as follows: *MC* or *VS* (low phonemic, low visual similarity); *FS* or *TD* (high phonemic, low visual similarity); *OQ* or *XY* (low phonemic, high visual similarity); *BP* or *MN* (high phonemic, high visual similarity) and *AB* (used for practice). Phonemically similar letter pairs shared a common vowel. Visually similar pairs shared a number of visual features (straight lines, curves, corners) while differing on only one or perhaps a few features. Though chosen independently, the classification

of letter-pairs in terms of visual similarity agrees very well with the visual confusion matrix published by Laughery and Harris (1970). The only disagreement is the pair *XY* which they regarded as dissimilar.

Subjects indicated their solution to each problem by ticking one of two spaces marked *true* and *false*. They were instructed to work rapidly whilst making as few errors as possible and were allowed to spend three minutes on each sheet. Each subject completed five sheets consisting of an initial practice sheet (using the letter pair *AB*) and then one sheet for each of the four combinations of phonemic and visual similarity of the letter pairs. Half of the subjects worked with the pairs *MC*, *FS*, *OQ* and *BP* while the remainder worked with *VS*, *TD*, *XY* and *MN*. All subjects were tested individually.

Design

A Latin Square was used to determine the order of presenting the four conditions. A three-way split-plot analysis of variance was carried out on the numbers of questions correctly solved. The within-subjects factors were phonemic and visual similarity. The between-subjects factor corresponded to the two sets of letter pairs.

Results

Mean numbers of correct solutions and errors are shown in Table VI. Data from the two sets of letter pairs have been combined since there were no effects attributable to this factor ($F < 1$ in all cases). Phonemic similarity reduced the number correctly solved by a small but significant amount ($F(1,30) = 10.47$, $P < 0.01$) while there was no effect of visual similarity ($F(1,30) < 1$) nor any interaction between the two types of similarity ($F(1,30) < 1$). Errors were relatively infrequent and did not appear to vary with either type of similarity. Thus performance on the verification task resembles the memory span in being sensitive to the effects of phonemic similarity. The absence of any evidence for visual coding is also reminiscent of the span task.

TABLE VI

Mean numbers of correct solutions and errors made on the reasoning task as a function of phonemic and visual similarity of the letter-pairs used in the problems

	Low phonemic similarity		High phonemic similarity	
	Low visual similarity	High visual similarity	Low visual similarity	High visual similarity
Mean no. of correct solutions	43.2	42.9	40.9	39.8
Mean no. of errors	1.8	0.9	0.9	1.5

General discussion

The working memory hypothesis

Two consequences of the hypothesis that sentence verification utilizes a working memory were tested: (i) additional short-term storage loads should adversely affect verification speed, and (ii) verification should make use of speech-coded information. The coding prediction was supported in Experiment IV where

verification speed was found to be reduced by phonemic similarity of the task material. Concerning the first, near-span STM *preloads* had no effect on verification speed (Experiment II, equal stress instructions) but *concurrent* STM loads of similar size had a large interfering effect (Experiment III). The results are therefore broadly consistent with the WM hypothesis, but the difference between the effects of the preload and concurrent load procedures was not entirely expected. The detailed effects can be evaluated against two quite different views of WM that were mentioned in the introduction. According to one, WM is simply a scratch-pad store of limited capacity. On the other it is a system with a limited capacity for executing control processes in addition to any storage function.

(a) *Working memory as a scratch-pad store*

If WM is a limited-capacity store interference between sentence verification and recall should occur when the combined storage demands of both tasks exceed the store's capacity. In Experiment I preloads of up to two items did not interfere with verification. However the delay of 1 s between presentation of the memory items and presentation of the verification task may have enabled subjects to consolidate the items prior to verification, using a strategy similar to that induced by memory stress instructions in Experiment II. Clearer evidence comes from the subjects given equal stress instructions in Experiment II who showed no interference from storing six additional letters during verification, even when all of them were recalled correctly. Since memory span for letters is between six and seven items (Cavanagh, 1972) it would appear that the short-term storage requirement of the verification task is minimal. A further test of the scratch-pad view follows from the reasonable assumption that verifying grammatically complex sentences ought to require more of the available scratch-pad storage space. Recall of the memory preload should therefore be poorer after a complex sentence than after a simple one (when any effects of delay on recall are controlled). Experiment II gave no evidence for such a relationship. Thus, so far as its role in sentence verification is concerned, the scratch-pad view of WM is unsupported by the present evidence.

(b) *Working memory as an executive*

The scratch-pad hypothesis has even further difficulty in explaining why concurrent storage loads produced interference while preloads had no effect. (The group for whom preloads did produce interference can be disregarded because of the particular strategy which they adopted.) The alternative is to regard the source of interference as an overload of information-processing capacity. A concurrent load can be seen to force the subject to divide his attention between the separate processes of verification and short-term storage, hence the interference in this situation. Furthermore, concurrent load interfered more severely with the verification of grammatically complex sentences while the simpler constructions showed much less interference. This is consistent with a processing-overload interpretation since complex sentences can be assumed to require more processing than simple ones.

With a preload the subject is not so constrained by the task demands. He is free to "dump" the memory items in short-term storage before verifying the sentence, returning to the store to retrieve items after making the verification response. The very rapid decline in recall accuracy with increasing verification latency observed in the preload experiments is consistent with the suggestion that little effort was made to process and therefore update the memory items during verification itself. Thus the preload procedure did not force division of processing capacity and there was no interference, even though the preload made near-span demands on the subject. The general conclusion is therefore that the storage demands of the verification task are much less important than its processing demands. If the concept of a working memory common to the span and a range of cognitive tasks is accepted, the present evidence suggests that it is a limited-capacity executive processing system rather than a scratch-pad store. The generality of this conclusion must depend on extending the present techniques to other tasks besides sentence verification.

(c) Other evidence concerning working memory

Baddeley and Hitch (1974) reported some preliminary experiments on the role of WM in comprehension of verbal material and free recall, in addition to a brief account of the present series of studies. Encouragingly, both these tasks behaved like sentence verification, showing interference from near-span concurrent memory loads and evidence of phonemic coding. Although providing some support for the generality of the working memory hypothesis, the results have not yet been subjected to the type of analysis which would have bearing on the scratch-pad vs. executive interpretations of WM. In the earlier report it was tentatively proposed that WM might act as a highly flexible limited-capacity system having both storage and executive capabilities. As has been seen, detailed analysis of the sentence verification task has not borne out this particular interpretation. However it remains an interesting possibility that may yet receive support in further studies of a wider range of tasks.

Articulation and WM

The sentence verification task showed small but reliable effects of phonemic similarity and suffered slight interference from concurrent articulation of a predictable sequence. These observations are compatible with the idea that the articulatory system is a separate part of the total information-handling complex, and that sentence verification places only a small load on this subsystem. The function of the articulatory system is presumably to allow temporary storage of a small number of items without the need for the more central WM to be unnecessarily encumbered. In non-verbal tasks it may be that the central WM is involved but the more peripheral articulatory system is not.

In summary, it is claimed that the limits of STM play a central role in human information processing. Verbal tasks at least seem likely to involve the interplay of two systems, the WM executive and the articulatory system. Though crude in its conclusions it would appear that the present approach has much to recommend

it. The memory span is an exceedingly well-studied task about which a great deal is known. Yet the role of the mechanisms underlying the span in more general situations is extremely poorly understood. Two-task studies in which one of the tasks is related to the memory span provide a direct and obvious way of exploring this question further.

We are grateful to the Medical Research Council for financial support, and thank Neil Thomson for useful discussion. We also wish to thank our reviewers, particularly Dr Alan Allport whose comments led to improvement in the final paper. The experiments were carried out while both authors were at the Laboratory of Experimental Psychology, University of Sussex and at the Department of Psychology, University of Stirling.

References

- ANDERSON, J. R. and BOWER, G. H. (1974). *Human Associative Memory*. New York: John Wiley.
- ATKINSON, R. C. and SHIFFRIN, R. M. (1971). The control of short-term memory. *Scientific American*, **225**, 82-90.
- BADDELEY, A. D. (1966). Short-term memory for word sequences as a function of acoustic, semantic and formal similarity. *Quarterly Journal of Experimental Psychology*, **18**, 362-6.
- BADDELEY, A. D. (1968). A three-minute reasoning test based on grammatical transformation. *Psychonomic Science*, **10**, 341-2.
- BADDELEY, A. D., DE FIGURED, J. W., HAWKSWELL-CURTIS, J. W. and WILLIAMS, A. N. (1968). Nitrogen narcosis and performance under water. *Ergonomics*, **11**, 157-64.
- BADDELEY, A. D. and HITCH, G. J. (1974). Working Memory. In BOWER, G. (Ed.), *The Psychology of Learning and Motivation: Advances in Research and Theory*, Vol. III. New York: Academic Press.
- BROADBENT, D. E. (1958). *Perception and Communication*. London: Pergamon Press.
- BROADBENT, D. E. (1971). *Decision and Stress*. London: Academic Press.
- BROWN, I. D., TICKNER, A. H. and SIMMONDS, D. C. V. (1969). Interference between concurrent tasks of driving and telephoning. *Journal of Applied Psychology*, **53**, 419-24.
- CAVANAGH, J. P. (1972). Relation between the immediate memory span and the memory search rate. *Psychological Review*, **79**, 525-30.
- CHASE, W. G. and CLARK, H. H. (1972). Mental operations in the comparison of sentences and pictures. In GREGG, L. W. (Ed.), *Cognition in Learning and Memory*. New York: John Wiley.
- CONRAD, R. (1962). An association between memory errors and errors due to acoustic masking of speech. *Nature*, **193**, 1314-5.
- CONRAD, R. and HULL, A. J. (1964). Information, acoustic confusion, and memory span. *British Journal of Psychology*, **55**, 429-32.
- CRAIK, F. I. M. (1971). Primary memory. *British Medical Bulletin*, **27**, 232-6.
- GLANZER, M. (1972). Storage mechanisms in recall. In BOWER, G. (Ed.), *The Psychology of Learning and Motivation: Advances in Research and Theory*, Vol. V. New York: Academic Press.
- GLANZER, M., KOPPENAAL, L. and NELSON, R. (1972). Effects of relations between words on short-term storage and long-term storage. *Journal of Verbal Learning and Verbal Behaviour*, **11**, 403-16.
- HAMMERTON, M. (1969). Interference between low information verbal output and a cognitive task. *Nature*, **222**, 196.
- KANARICK, A. F. and PETERSON, R. C. (1969). Effects of value on the monitoring of multi-channel displays. *Human Factors*, **11**, 313-20.

- LAUGHERY, K. R. and HARRIS, G. J. (1970). Visual and auditory intrusion errors in short-term memory. *Journal of Experimental Psychology*, **83**, 101-6.
- LEVINE, M. (1966). Hypothesis behaviour by humans during discrimination learning. *Journal of Experimental Psychology*, **71**, 331-8.
- MILLER, G. A. (1956). The magical number seven, plus or minus two. *Psychological Review*, **63**, 81-7.
- MURRAY, D. J. (1967). The role of speech responses in short-term memory. *Canadian Journal of Psychology*, **21**, 263-76.
- NEWELL, A. and SIMON, H. A. (1972). *Human problem solving*. New York: Prentice-Hall.
- PETERSON, L. R. (1969). Concurrent verbal activity. *Psychological Review*, **76**, 376-86.
- POSNER, M. I. (1965). Memory and thought in human intelligence. *British Journal of Psychology*, **56**, 197-215.
- SHIFFRIN, R. M. (1970). Memory search. In NORMAN, D. A. (Ed.), *Models of Human Memory*. New York: Academic Press.
- TRABASSO, T. (1972). Mental operations in language comprehension. In FREEDLE, R. O. and CARROLL, J. B. (Eds.), *Language Comprehension and the Acquisition of Knowledge*. Washington D.C.: Winston.
- WASON, P. C. and JOHNSON-LAIRD, P. N. (1972). *Psychology of Reasoning: Structure and Content*. London: Batsford.
- WELFORD, A. T. (1968). *Fundamentals of Skill*. London: Methuen.
- WICKELGREN, W. A. (1965). Short-term memory for phonemically similar lists. *American Journal of Psychology*, **78**, 567-74.

Revised manuscript received 27 October 1975